

Energy Modes

Control Theories in Microgrid Applications



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1 Introduction

The electricity demands of the human population is increasing rapidly due to steadily increasing world population [1] and rising share of people with access to modern technology. Van Vuuren predicts that by 2050 the energy demand will be up to 790 exajoule, a large increase compared to 580 exajoules used in the year 2023 [2].

This significant increase in demand comes at a time the world is actively dealing with and attempting to combat the effects of global warming, an issue that will define the 21st century. To meet the increasing energy demand the energy infrastructure will need to be upgraded and expanded, however it is important that it is done in a sustainable manner with global warming and the future of our planet in mind.

A crucial component in addressing this growing energy demand while mitigating the impact of global warming lies in the transition to renewable energy sources. Unlike traditional fossil fuels, renewable energy sources offer a more sustainable path forward, with significantly less impact on global warming. As private and public actors shift focus toward energy sources with reduced environmental impact, renewable energy sources are predicted to play a significant role in global energy production. A prediction for the development of energy consumption can be seen in Table 1.

However, building and integrating renewable energy sources is not without challenges. The energy output from renewable energy sources is less stable compared to traditional power plants. The fluctuations are a product of varied wind and sunshine amount. Also, by the nature of renewable energy sources, the power plants are often decentralized and built in remote locations, often far away from traditional energy infrastructure.

The concept of microgrids is becoming popular as a way to deal with the aforementioned challenges. Microgrids are localized, self-sufficient systems that can operate independently or in conjunction with the larger grid. However, to integrate a microgrid into the larger grid one must make sure that voltage and similar parameters are synchronized. This report investigates the mathematics and model describing the control issues at play [3].

1.1 Definitions

A microgrid is a localized energy system that can operate independently or in conjunction with the larger energy grid. It often consists of generators utilizing renewable energy sources such as, wind turbines or solar panels, energy storage systems and control mechanisms to regulate energy production and voltage.

Renewable energy sources are sources of energy that are not naturally depleted by energy generation. Some examples are water, wind, solar and biofuel/biomass. In contrast with energy sources that do get depleted with use such as coal and oil. A significant benefit of renewable energy sources compared to their non renewable counterpart is the lower environmental impact.

2 Background

The increased energy demand of the global economy is evident and tied in large parts to the economical growth. An increase of 1% in GDP is shown to increase the energy usage by 1.3%. While the demand for energy will not dissipate, the need to use sustainable energy sources is a must. Since it is estimated that the supply of oil and other fossil fuels will become more scarce towards the second half of this century. Looking at the energy production from 2016 as shown in Figure 1, the shares of energy production different energy sources hold. Renewable energy accounts for roughly 25% of the total energy production while around 50% is accounted for by non-renewable energy such as fossil fuel. Energy production from renewable energy has increased considerably between the years of 2008 and 2016 as energy generation from solar power and wind power increased with 280 000 MW and 460 000 MW respectively, constituting an increase of 1 950% for solar and 400% for wind. [3]

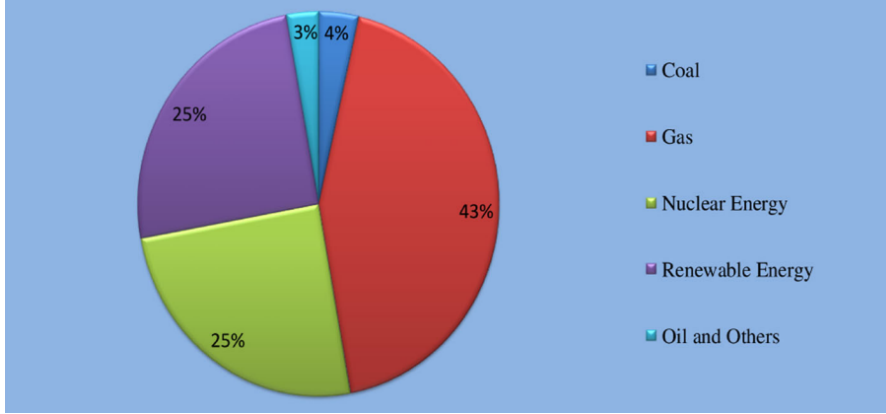


Figure 1: Percentage of Electric Power Produced from Various Energy Sources in 2016 [3].

While the increases are large they only helped to bring renewables energy share up to 25%. Using the future predictions for the energy productions in table 1, fossil fuels will remain stagnant while renewable sources increase drastically between 2016 and 2100. This however is the primary energy production and not the net energy production. There is criticism to these predictions in the fact that they do not take into account the added energy cost of production and transport, while also neglecting the effect that the energies have on the environment. [3, 2]

However, there isn't only an environment downside to fossil fuel but also a grave social cost. The anthropogenic air pollution from burning fossil fuel has lead to death of 3.6 million people around the world each year. [4]

Fuel	2016 (Actual)	2040 (EIA)	2040 (ExxonMobil)	2040 (OPEC)	2050 (van Vuuren)	2100 (van Vuuren)
FF	475	591	575	583	460	482
RE	80	135	195	188	280	408
Nuclear	25	40	54	77	50	180
Total	580	766	824	848	790	1070

Table 1: Primary energy production projections by fuel type (in exajoule [$10^{18}J$]), 2016-2100. Taken from [2]

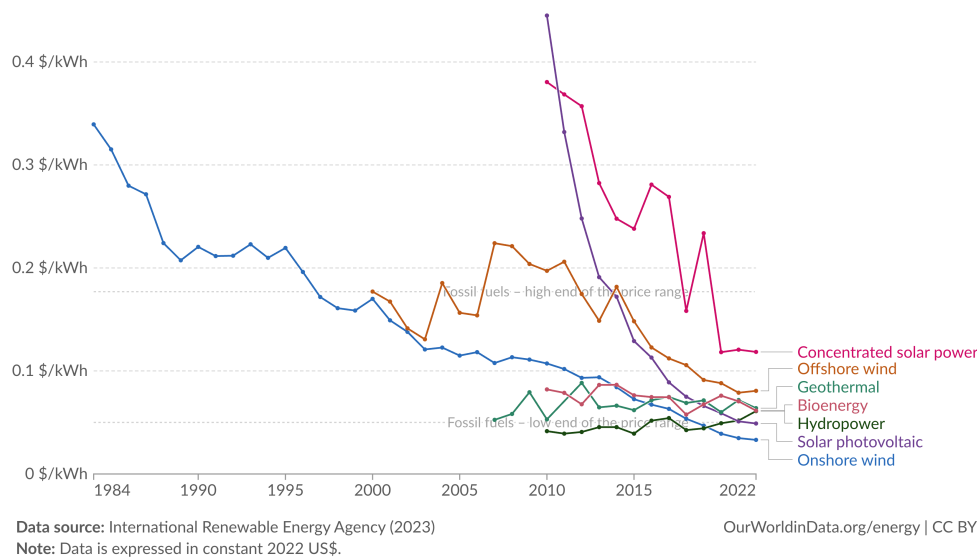
2.1 Cost of transition

The cost of transitioning from fossil fuels to renewable energy sources is an expensive process that depends on multiple variables such as scale, technology, grid infrastructure, and more.

The improvement and growth of investment in renewable technologies over the past decades have led to significant reductions in both the upfront and lifetime costs of renewable energy sources, such as solar and wind. This decline is largely attributed to technological advancements, economies of scale, and increased production efficiency, making renewables more cost-competitive with traditional fossil fuels, see figure 2. For instance, the price of solar modules has fallen by over 80% from 2010 to 2022. 5

Levelized cost of energy by technology, World

The average cost per unit of energy generated across the lifetime of a new power plant. This data is expressed in US dollars per kilowatt-hour¹. It is adjusted for inflation but does not account for differences in the cost of living between countries.



1. Watt-hour: A watt-hour is the energy delivered by one watt of power for one hour. Since one watt is equivalent to one joule per second, a watt-hour is equivalent to 3600 joules of energy. Metric prefixes are used for multiples of the unit, usually: - kilowatt-hours (kWh), or a thousand watt-hours. - Megawatt-hours (MWh), or a million watt-hours. - Gigawatt-hours (GWh), or a billion watt-hours. - Terawatt-hours (TWh), or a trillion watt-hours.

Figure 2: The average cost per unit of energy generated across the life of a new power plant from 1984 to 2022 6.

While the costs of renewable energy have decreased dramatically, integrating these sources into existing grid infrastructure presents significant challenges. Traditional fossil fuel plants operate on a central grid model, similar to a heart pumping blood through the body, while renewables rely on a distributed model with less stable energy production. Thus, many parts of the grid must be reconfigured, and larger energy storage options, such as batteries, are required to handle the varying load and stabilize the grid. 3, 7

2.2 Advantage of Microgrids

Microgrids play a crucial role in the energy transition by enhancing the integration and use of renewable energy sources, such as solar and wind, while reducing dependency on fossil fuels. They provide localized electricity generation and storage, allowing for energy resilience and reliability, especially when disconnected from the main grid (island mode). Microgrids support energy independence and flexibility by stabilizing power from intermittent renewable sources and ensuring continuous supply through energy storage solutions. This adaptability makes mi-

microgrids essential in balancing fluctuating renewable energy outputs and improving the overall power quality and sustainability of energy systems. Despite challenges like voltage oscillations and technical complexities, microgrids are vital for a sustainable future as they enable higher renewable energy penetration, reduced transmission losses, and localized power management, addressing both environmental and energy security concerns. [3]

A practical example of how microgrids are supporting the renewable energy transition is the Siemens project funded by the U.S. Department of Energy (DOE). This project developed a three-layer energy management system for microgrids, enabling them to operate autonomously during power outages or emergencies. The system allows microgrids to forecast energy availability, balance loads, and cooperate with other microgrids in case of disruptions like cyberattacks. Additionally, it can autonomously restore power using renewable energy sources such as solar or wind, significantly reducing downtime. This makes microgrids a critical tool for ensuring energy reliability as climate-related disruptions become more frequent. [8]

3 Model and control methods

Microgrids are characterized by being fully autonomous: the fact that they could work as single entities, separated from the main grid, means that they have to be able to carry out every necessary task for their correct functioning by themselves.

One of such important tasks is the control one, which has to manage the deviations of voltage and frequency that could arise due to the variations of renewable energy sources.

To deal with this issue, many control methods have been proposed, and the following sections will focus on two of them, namely the ILQG and Proportional integral derivative controller (PID-controller).

3.1 Integral Linear Quadratic Gaussian (ILQG) Controller

The Integral Linear Quadratic Gaussian (ILQG) controller is an advanced control algorithm used for tracking performance of single-phase islanded microgrids. It provides high bandwidth, large gain and significant phase margin. By introducing an integral action to the LQG design, another well known state-space algorithm, the ILQG controller achieves a better tracing performance of reference signals, while maintaining robustness against uncertainties. [3]

3.1.1 Structure of LQG

The design of the ILQG controller is an updated version of the LQG controller. As such, it is fundamental to introduce the structure of an LQG controller, which begins with a state-space model of the plant, as follows:

$$\dot{x}(t) = Ax(t) + Bu(t) + D_1\alpha_1(t) \quad (1)$$

$$y(t) = Cx(t) + D_2\alpha_2(t) \quad (2)$$

Where:

- x is the state vector. It represents the internal variables that describe the current condition of the microgrid system. An example of it can be the following:

$$x(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{pmatrix} = \begin{pmatrix} \text{Voltage of battery} \\ \text{Power output from an inverter} \\ \text{Frequency deviation} \\ \text{State of charge of battery} \end{pmatrix}$$

- y is the output of the plant. It represents the measurable quantities in the system, variables that can be regulated or monitored. An example of it can be the following:

$$y(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{pmatrix} = \begin{pmatrix} \text{Voltage at a critical node} \\ \text{System frequency} \\ \text{Loads power demand} \end{pmatrix}$$

- u is the control input,
- A is the system matrix,
- B is the input matrix,
- C is the output matrix,
- D_1 and D_2 represent the noise matrices of the system,
- α_1 and α_2 represent the process noise.

The goal of the LQG controller is to obtain a law for the control inputs dependent on the state of the system:

$$u(t) = Kx(t). \quad (3)$$

However, while this controller is effective in tracking transient errors, it is unable to account for steady-state errors. This means that, with an LQG controller, the error can accumulate over time.

As an example, imagine that the voltage at a particular node in the microgrid is supposed to be regulated at 230V. Moreover, let the microgrid experience a constant load disturbance that consistently draws more power than anticipated, causing the voltage at this node to drop slightly, say to 228V. The voltage error is then 2V below the desired setpoint, and this error persists over time due to the constant disturbance. In such a context, the LQG controller would likely not be able to bring the voltage back to the desired 230V, because it does not incorporate an integral action that would accumulate the 2V error over time. Instead, it might stabilize the voltage around 228V.

3.1.2 Structure of ILQG

The ILQG controller, on the other hand, strives to design a system to be robust against uncertainties. In fact, let $r(t)$ be the desired output vector and $\dot{e}(t) = |r(t) - y(t)|$ be the output error vector, with the ILQG method the following is obtained:

$$u(t) = \bar{K}\bar{x}(t), \quad (4)$$

where $\bar{x} = \begin{pmatrix} x \\ e \end{pmatrix}$ is the augmented state vector of the system, that clearly depends on the accumulated error.

Therefore, to implement the ILQG controller an extended version of the state-space model (1) and (2) is needed (the noise terms are set to zero for simplicity's sake):

$$\dot{\bar{x}}(t) = \bar{A}\bar{x}(t) + \bar{B}u(t) + \bar{I}e \quad (5)$$

$$\bar{y}(t) = \bar{C}\bar{x}(t) \quad (6)$$

where $\bar{A} = \begin{pmatrix} A & 0 \\ -C & 0 \end{pmatrix}$, $\bar{B} = \begin{pmatrix} B \\ 0 \end{pmatrix}$, $\bar{C} = (C \ 0)$ and $\bar{I} = \begin{pmatrix} 0 \\ I \end{pmatrix}$ are the augmented matrices of the initial system.

In this upgraded framework, the goal of the ILQG method is to minimize the cost function

$$J = \int_0^{+\infty} \bar{x}^T \bar{Q} \bar{x} + u^T R u \, dt, \quad (7)$$

where $\bar{Q} \geq 0$ is the extended state control matrix and $R > 0$ is the control cost matrix: these are weight matrices that assign at each state or control a certain importance. That is, if a specific state $\bar{x}_i$ is more important to control, the corresponding value q_{ii} needs to be increased. In other words, a high value in $\bar{Q}$ means that the controller will react strongly to deviations in the associated state, while a low value will allow more deviations to happen. For R , on the other hand, the higher values, the more conservative the controller will be with the control inputs $u(t)$, resulting in slower system response, but also lower control effort.

Therefore, solving (7) means balancing $\bar{Q}$ and R .

$\bar{K}$ is then obtained solving the following Algebraic Riccati Equation:

$$\bar{A}^T P + P \bar{A} - P \bar{B} R^{-1} \bar{B}^T P + \bar{Q} = 0, \quad (8)$$

where P is the solution of such equation, and $\bar{K}$ is

$$\bar{K} = R^{-1} \bar{B}^T P. \quad (9)$$

3.2 PID-controller

The PID controller is one of the most widely implemented control systems in various industries, including the implementation in microgrids. It offers a way simple way to design a system whilst also getting great performance. It is highly valued for its ability to handle systems with lower order, transfer functions, and given minimal knowledge of the underlying system, it offers a way to minimize the steady-state error.

3.2.1 Structure

The PID controller consists of three parts, each corresponding to the letters of the abbreviation.

- K_p is the proportional gain,
- K_i is the integral gain, and
- K_d is the derivative gain.

Through these components we are able to modify the systems dynamics to correctly respond to changes. However, to get the desired performance and responses from the system we have to tune our previously stated parameters. This can be done by using different tools such as:

- Root Locus Method,
- Ziegler-Nichols Method,
- Cohen-Coon Method,
- Chien-Hrones-Reswick Method,
- Wang-Juang-Chan Method.

Among these methods, the Cohen-coon method stands out in its suitability to different processes. This method offers a self-regulating procedure that automatically tunes the PID parameters, leading to improved performance in controlling various systems. [3]

3.2.2 Cohen-Coon Tuning Formula

The Cohen-Coon tuning rules for the PID controller are as follows:

$$K_p = \frac{1.35}{a} \left(1 + \frac{0.18\tau}{1 - \tau} \right)$$

$$K_i = \frac{K_p(1 - 0.39\tau)}{L(2.5 - 2\tau)}$$

$$K_d = \frac{K_p L(0.37 - 0.37\tau)}{1 - 0.81\tau}$$

Where:

- L is the delay time,
- T is the time constant,
- $a = \frac{kL}{T}$, and
- $\tau = \frac{L}{L+T}$.

This formulation allows for greater stability in the controller and also better performance overall from the system.

3.2.3 Microgrid applications

The PID controller, in the context of microgrids, plays an essential role in voltage stability and the maintenance of consistent power delivery, even with fluctuating loads and disturbances. By using the Cohen-Coon method a PID controller can be made adaptive and dynamic to conditions, such as varying loads and noise disturbances, without manual adjustments. [9] This enables robust voltage tracking and more stable systems.

The ability to provide reliable voltage regulation in microgrids makes the PID controller valuable in the view of controlling and optimizing distributed energy systems.

3.3 Comparison

Table 2: Comparison between the ILQG and PID controllers [3]

Controller	Advantages	Limitations
Integral LQG Controller	High bandwidth, gain, and phase margin	
PID Controller	Low-order transfer function matrices, easy to implement	Low bandwidth

The ILQG controller provides advantages such as high bandwidth, gain and phase margin. This makes it suitable for systems requiring precise and robust control. Particularly, in the presence of noise and uncertainties this controller becomes well adapted. It excels in applications such as microgrids but involves a more complex design and implementation.

The PID controller in contrast, is widely used for its simplicity, relying on lower order transfer functions. The addition of being easy to implement makes it appealing. However this appeal is overruled by its limited performance in fast changing environments and lower bandwidth. While PID controllers are simpler, they do not offer as good of a performance as ILQG controller for complex systems.

4 Discussion

In this paper we have presented an important tool to combat climate change, namely the microgrid, and the challenges they face in relation to control theory. We predict that the use of microgrids will increase in tandem with the production increase of energy coming from renewable sources.

A drawback of the proposed ILQG model is its model-dependence and computational complexity, while, on the other hand, the PID controller simply reacts to the error between desired and actual output.

The ILQG controller, however more difficult to implement, does offer many advantages compared to the PID controller. The responsible parties need to weigh the advantages and drawbacks carefully when choosing an appropriate control theory model for each microgrid.

For future work one could attempt to implement the PID and ILQG controllers and compare their performance. Also optimizing the matrices for a particular microgrid to maximize performance is important for each and every microgrid.

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Energy Modes

Control Theories in Microgrid Applications

Oguzhan Akan, Xin Chen, Lorenzo Demaria, Alexander Hallberg, Anton Wärnberg

Introduction



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Introduction



- Energy consumption is predicted to increase by 36% from 2023 to 2050.
- Energy production from fossil fuels has a negative impact on climate change.
- Renewable energy sources are often dependent on variables with high variance.
- Often decentralized and built in remote locations.
- A microgrid is a localized energy system with the capability of independent work or in conjunction with the larger grid.



Background

Energy and Economic Growth

- Energy demand grows alongside GDP.
- Fossil fuel resources is predicted to become more scarce by the mid-21st century.

Renewable Energy Growth

- Solar and wind power generation increased drastically between 2008-2016.





Renewable Energy Projections

Energy Production Forecast (2016-2100)

- Production from fossil fuels show nominal growth.
- Renewable energy sees significant growth.

Fuel	2016 (Actual)	2040 (EIA)	2040 (ExxonMobil)	2040 (OPEC)	2050 (van Vuuren)	2100 (van Vuuren)
FF	475	591	575	583	460	482
RE	80	135	195	188	280	408
Nuclear	25	40	54	77	50	180
Total	580	766	824	848	790	1070

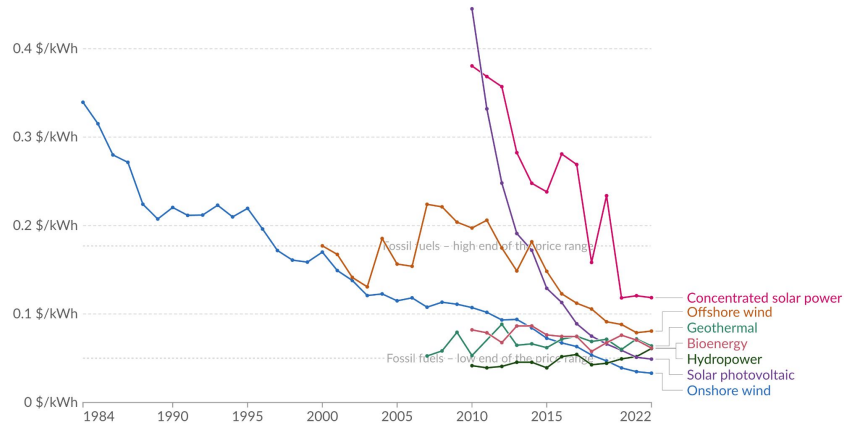
Table 1: Primary energy production projections by fuel type (in exajoule [$10^{18}J$]), 2016-2100.
Taken from [2]

Cost of Transition to Renewable Energy

- **High Initial Costs:** Transitioning from fossil fuels to renewable energy sources is expensive and depends on factors like scale, technology, and grid infrastructure.
- **Grid Reconfiguration:** Traditional fossil fuel grids are centralized
Renewable energy grids are more distributed
- **Cost Reductions:** Renewable technologies is much cheaper now

Levelized cost of energy by technology, World

The average cost per unit of energy generated across the lifetime of a new power plant. This data is expressed in US dollars per kilowatt-hour¹. It is adjusted for inflation but does not account for differences in the cost of living between countries.



Data source: International Renewable Energy Agency (2023)

Note: Data is expressed in constant 2022 US\$.

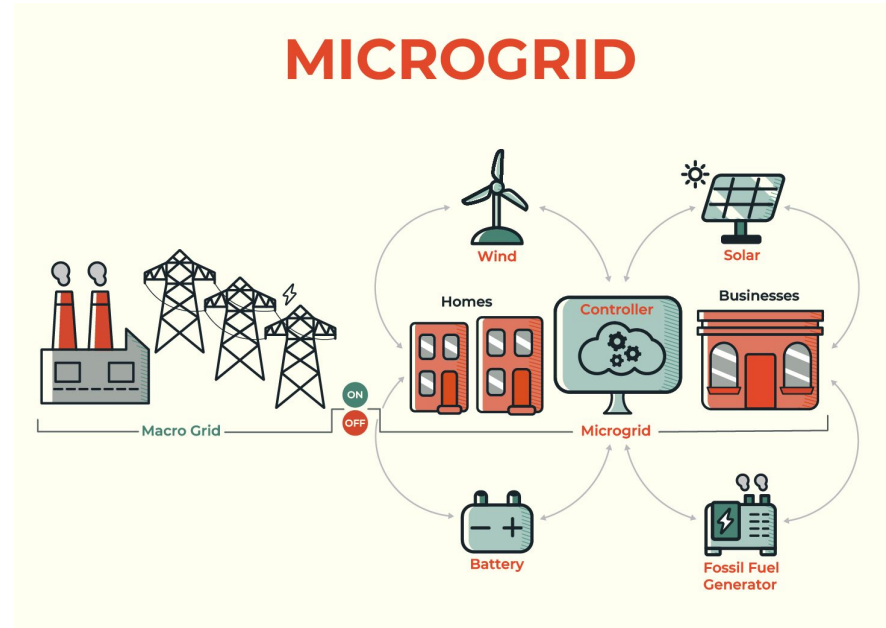
OurWorldinData.org/energy | CC BY

Advantages of Microgrids

Localized Power Generation: Microgrids generate and store energy from renewables like solar and wind, reducing reliance on fossil fuels.

Energy Resilience: Can operate independently during outages, providing reliable power in emergencies.

Grid Stability: Help balance fluctuations in renewable energy output, improving power quality and reducing transmission losses.





Model and control methods

- Microgrids are autonomous systems: need of control methods
- 2 examples: Integral Linear Quadratic Gaussian controller (ILQG) and Proportional Integral Derivative controller (PID)



ILQG

- State-space model required, similar to the LQG one

- LQG

$$\dot{x}(t) = Ax(t) + Bu(t) + D_1\alpha_1(t)$$

$$y(t) = Cx(t) + D_2\alpha_2(t)$$

where x is the state vector, y is the output of the plant, u is the control input

- ILQG

$$\dot{\bar{x}}(t) = \bar{A}\bar{x}(t) + \bar{B}u(t) + \bar{I}e$$

$$\bar{y}(t) = \bar{C}\bar{x}(t)$$

where $\dot{e}(t) = |y(t) - r(t)|$ is the error (r expected output), $\bar{x} = \begin{pmatrix} x \\ e \end{pmatrix}$ is the augmented state vector of the system



ILQG

- The goal is to obtain a control equation dependent on the state of the system

$$u(t) = \bar{K}\bar{x}(t)$$

- This is done minimizing the cost function J

$$J = \int_0^{+\infty} \bar{x}^T \bar{Q} \bar{x} + u^T R u \, dt$$

where $\bar{Q} \succeq 0$ is the extended state control matrix and $R > 0$ is the control cost matrix

- And then solving the Algebraic Riccati Equation

$$\bar{A}^T P + P \bar{A} - P \bar{B} R^{-1} \bar{B}^T P + \bar{Q} = 0$$

where P is the solution of such equation, and $\bar{K} = R^{-1} \bar{B}^T P$



PID

- **Widely Implemented Control System**
 - Simple design with excellent performance
 - Minimizes steady-state error with minimal system knowledge
- **Structure of PID Controller**
 - **Proportional Gain (K_p):** Responds to current error
 - **Integral Gain (K_i):** Eliminates steady-state error by integrating past errors
 - **Derivative Gain (K_d):** Predicts future error based on the rate of change



PID

- **Applications in Microgrids**
 - Maintains voltage stability and consistent power delivery
 - Adapts to fluctuating loads and disturbances without manual adjustments
- **Importance in Distributed Energy Systems**
 - Enables robust voltage tracking
 - Valuable for controlling and optimizing microgrids



Comparison

- **Precision vs. Simplicity:** ILQG is more precise but complex. PID is easier to implement but less adaptable.
- **Performance:** ILQG handles fast system changes better, while PID is limited by lower bandwidth.

Final remarks

- More energy will need to be produced and produced in a sustainable manner.
- The microgrid structure is not uncommon for renewable energy sources, however it has a number of control theoretical challenges.
- We presented the ILQG and PID controllers, both with their own pros and cons.
- Let us all work towards a sustainable society 🌲🖤💛



Thank you for listening!